AI-Use Appendix (one page)

Scope and tools

This draft was prepared with the OpenAI Codex assistant in the desktop app. The exact underlying model identifier was not independently exposed and is not claimed. Tools actually used: Python 3 with NumPy for RK4, eigenvalues and checks; ReportLab for typesetting/vector figures; pdfplumber and PDF rendering for inspection; web search for limited primary-source comparison. The student executed the supplied script in MATLAB; output CSVs, plots and a completion screenshot were reviewed. The MATLAB release was not recorded. No symbolic algebra package or Octave run was used.

Prompt and use

User prompt: “Use the uploaded ME319 Assignment 1 PDF to help complete a rigorous, original steam-engine governor project. Develop and check an historically plausible novel concept, derive and verify control equations and stability, implement reproducible MATLAB/Octave simulations and plots for ±30% load, and prepare a maximum-six-page IEEE-style paper plus one-page honest AI-use appendix, repository-ready code and README. Clearly mark assumptions and anything requiring the student’s own validation; do not fabricate originality verification, simulation results, citations, or GitHub URL.” The assistant retrieved the assignment, proposed the mechanism, derived the model, wrote source files and drafted this report.

Accepted provisionally

Conservation-of-volume inventory integration; differential-pressure proportional sensing; negative feedback sign; cubic characteristic polynomial; Routh condition; reproducible RK4 and analytic numerical checks. Engineering reasons: inventory preserves load-dependent throttle position at zero speed error, while proportional correction and valve damping avoid an undamped pure-integral loop. Selected gains satisfy the cubic stability test. The student executed the simulations and supplied the resulting files. Independent rederivation and physical validation remain outstanding.

Modified or rejected during preparation

A fixed tank outlet was rejected as a basis for ideal integration because its flow varies with level. The proposal instead requires a translating constant-head outlet, whose feasibility remains unverified. Broad claims of novel hydraulic regulation were rejected after comparison with Maxwell/Siemens prior art. Initially unsupported execution claims were withheld; MATLAB execution is now documented by student-supplied outputs. Measured parameters, proven historical construction, complete physical validation is still not claimed. No experimental results were generated.

Validation actually completed

Independent Python numerical integration agreed with an eigenmode solution within 2.26e-11 state units; halving the time step changed states by less than 2.12e-11. Opposite load steps had exact perturbation symmetry. A pressure-sensor nonlinear comparison and 64 parameter-corner check were executed. The student also ran MATLAB. Both cases gave a 1.89 s peak time and 10.43 s settling time; reported metrics matched Python within 1e-9, and the maximum MATLAB RK4/expm error was 2.251e-11. These checks test the assumed equations only, not physical feasibility or originality. The PDF was checked for page count and rendered layout.

Student / team actions still required

MATLAB execution and output comparison are complete. Independently rederive equations; verify component chronology and perform broader prior-art comparison; dimension the lever and verify float/valve forces; calibrate parameters and test omitted dynamics. Author details have been added: vikas H R, 23B2282. The repository is https://github.com/vikashr-alt/steam-governor. Update this appendix with actual student changes and tools, and submit honest confidential peer evaluations. No contribution percentages, experimental validation or complete student engineering review are implied by this draft. This appendix must be revised if the final project differs.